

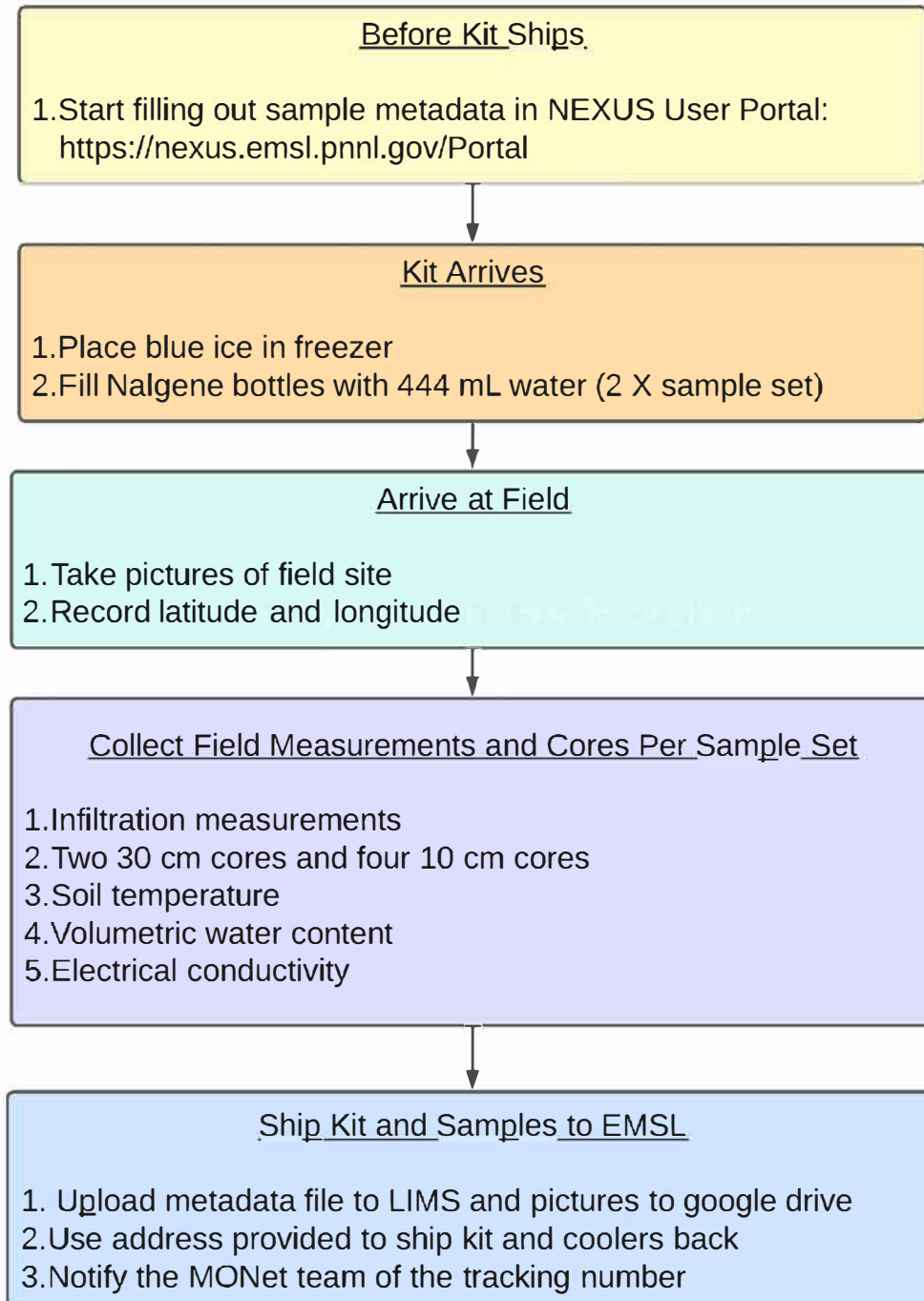
MONet Field Sampling Protocol

This protocol describes soil core sampling for the Molecular Observation Network (MONet). Use of any other sampling materials or protocols, sampling at other than the locations identified in the Participant's MONet sample collection proposal(s), or sampling prior to the proposal being approved by MONet, is outside the scope of the MONet project. Data will be publicly available according to EMSL's data policy. **Sampling should be completed Monday-Tuesday and shipped "First Overnight" using the PNNL FedEx account number.** Please contact monet@pnnl.gov with any questions.

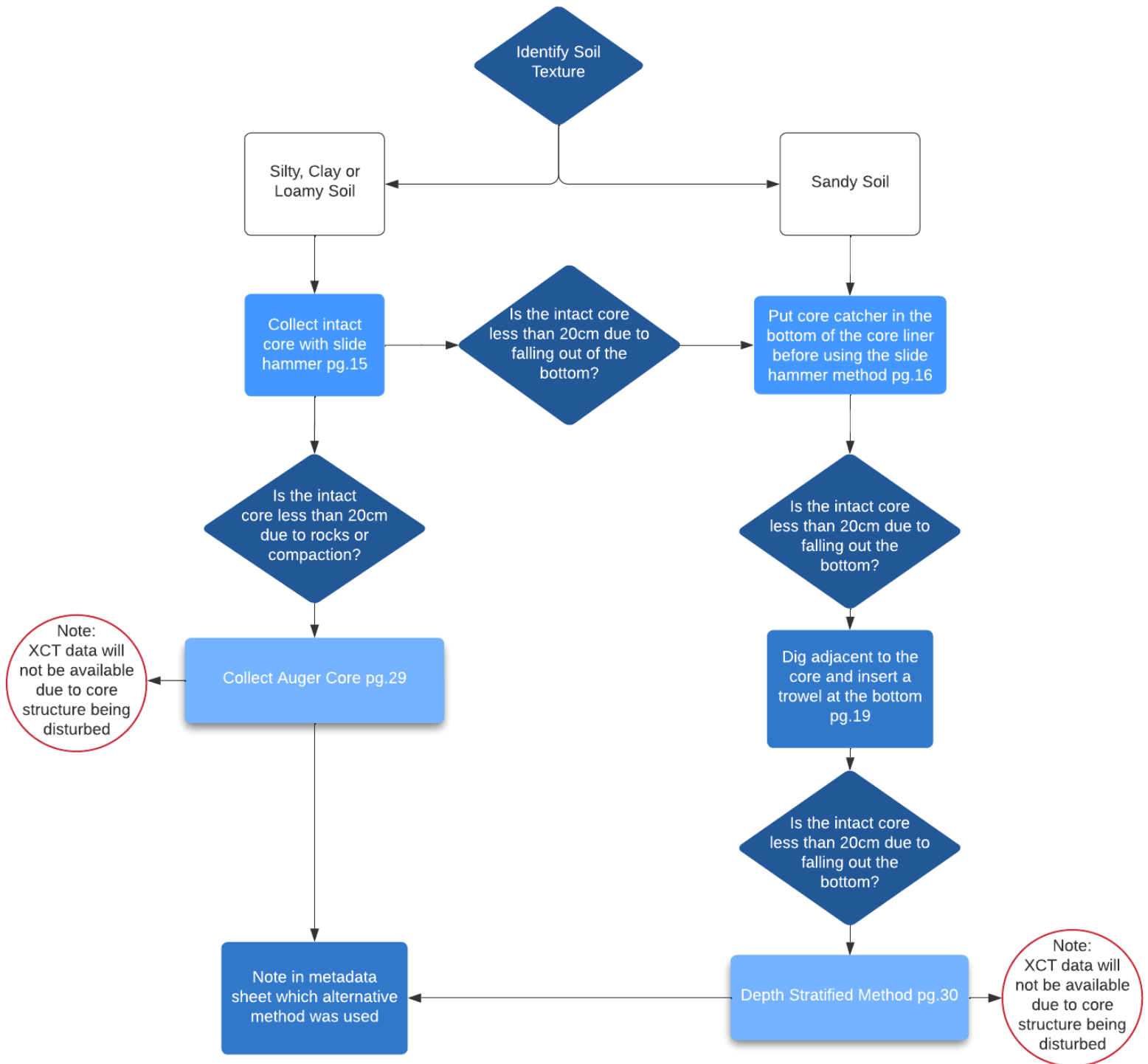


U.S. DEPARTMENT OF
ENERGY

MONet Field Sampling Overview



Decision Flowchart



Prior to Arriving at the Field for Sample Collection

- 1) Start filling out site metadata within your NEXUS workspace (<https://nexus.emsl.pnnl.gov/Portal>).
- 2) Ensure that the kit contains all items listed in the materials section above. Contact monet@pnnl.gov if you are missing anything.
- 3) Download the ZENTRA Utility App from the app store or by scanning the QR code from the meter case.



- 4) Place the blue ice provided for shipping in a freezer.
- 5) Fill the each of the four included bottles with 444 ml of tap water for infiltration measurements.

Sampling Tips and Tricks

Before Sampling

- Check to ensure all supplies are included in the kit and coolers. A list and diagram of necessary equipment is on page 6 of the sampling protocol.
- Check that the included sticker labels (in clipboard) are correct & correspond with your sampling sites.
- Note any pre-existing damage on equipment.
- Place provided cold packs in the freezer.

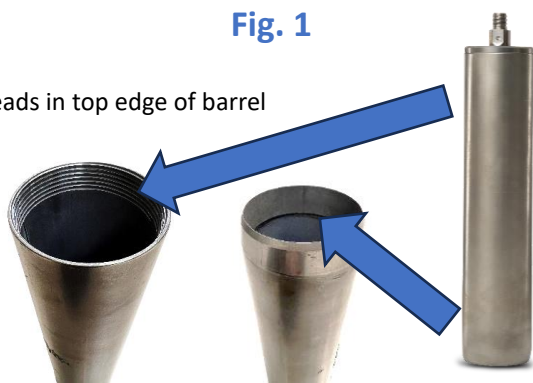
During Sampling

Use caution – edge of coring barrel is sharp!

- Place sticker labels on the core liners before inserting the liner into the barrel.
- For best coring results (Fig.1):
 - o Use the wire brush to clean the threads on the inside of coring barrel and cap in between **every** core.
 - o Use the wire brush to clean the bottom inner lip of the barrel that the core liner sits on.
- **Ensure all pieces of the coring equipment are fully secured together, hand tight.** Do not overtighten. (Fig. 2)
- To remove the coring barrel from the ground, rock the barrel back and forth to loosen from surrounding soil.

Fig. 1

Threads in top edge of barrel



Inner lip in bottom edge of barrel

Fig. 2



Correct

Incorrect

Use caution, sharp edges!

After Sampling

- **Clean and dry** all equipment used before shipping the kit back.
- Ensure all samples are properly labelled.
- Place frozen cold packs in with the samples to maintain temperature.
- Notify lab staff via email at monet@pnnl.gov with the following:
 - o When samples have been shipped & provide tracking number.
 - o If all samples collected are full 30cm and 10cm cores.
 - o Any damage/issues that occurred with the equipment while sampling.

Other Notes:

- The fixed wrench only fits the auger cap, use the adjustable wrenches on the regular coring barrel.
- Reach out to lab staff via email at monet@pnnl.gov with any issues/questions while sampling.

Table of Contents

Materials	6
Metadata	8
Infiltration Measurements	9
Collect Intact Cores	15
Collect Field Measurements	23
Collect Surface Cores	29
Alternative Methods	31
Post Sample Collection.....	33

MONet Field Sampling Protocol

Sampling kit should contain the following:	Items provided by you:
1. Cooler a. Two Freezer blue ice packs (please place in freezer for minimum 48 hours prior to return shipping) b. Nitrile gloves (bags containing small, medium, and large gloves) c. 2 bags per site core to prevent leaking d. Zip top plastic storage bags e. Core liners (Two – 7.6 x 30 cm and four – 5 x 10 cm liners with caps per site) f. Four 7.6 cm core catchers (if needed) g. 1x 30cm pool noodle (used to stabilize cores/fill open space in coring tubes during shipping). h. Kimwipes (bag containing 20 wipes) 2. Pelican Case a. Clipboard i. Hard copy of data sheet ii. Writing utensils (2x pen, 1x marker) iii. Electrical tape iv. Ear protection v. Tape measure vi. Utility knife (w/extra blades) vii. Cleaning brush b. AMS slide hammer c. AMS Intact Corer with screwable cap d. AMS Soil Recovery Auger Corer (SRA) e. AMS Extender (60 cm and 905 cm) f. AMS Auger Handle g. Wrenches (fixed end and crescent) h. Strap wrench i. Black paddle j. Rubber mallet k. Infiltration ring l. Plastic wrap m. Four 500 ml Nalgene bottles (inside core barrels) n. Meter Group ZSC Bluetooth sensor + Teros 12 meter (w/extra batteries) o. 2x stainless-steel bench scraper p. Collapsible Shovel	1. Freezer blue ice packs (separate from shipping blue ice) to keep samples cold in the field 2. Method for collecting latitude and longitude in decimal degrees in the field (this can be done using a smart phone) 3. Method for taking pictures of field site and sample cores (this can be done using a smart phone) 4. Water for infiltration measurements (~1 L per site) 5. -20 C freezer to freeze ice packs prior to shipping (do not freeze cores) 6. Refrigerator to store cores prior to shipping 7. Access to a FedEx shipping location to return packages 8. Packing tape to close shipping kit

Materials

Bottom



Middle



Top



Collect Metadata

1. Record longitude and latitude in decimal degrees (smart phone app 'My GPS Coordinates') and the date and time of sampling.
2. For each sample set, take one picture of the sample area with the measuring tape extended to 1 meter and 1 additional picture of the landscape.
3. **Upload to MONet Google drive as a .jpg files after sampling** with the following naming format: "project number - sample set - measure/landscape"



61555_1_measure



61555_1_landscape

Infiltration Measurements

1. Wear a pair of nitrile gloves provided in the kit.
2. Clear the sample area of debris (aka litter) and place the infiltration ring on the soils.



Ensure you do not disrupt the first layer of soil.

3. Use the black paddle and mallet to drive the infiltration ring 2.5 cm into the ground.



If you run into an obstacle such as a rock, move the infiltration ring over to avoid the obstruction.



The infiltration ring did not enter the soil evenly, due to a rock. The ring was moved slightly to avoid the obstruction.

6. Pour 444 mL water into the ring lined with plastic wrap.



7. Remove the wrap leaving just the water in the ring and start recording the time using a stopwatch. Record the amount of time (in minutes) it takes for the water to infiltrate the soil. Stop timing when the soil surface is just glistening (see photos attached on the next page).



8. In the same ring at the same location, repeat steps 2-4 for a second infiltration measurement.



If infiltration takes more than 1 hour, abort measurement.



There is too much water in the infiltration ring, continue to wait before stopping the timer.



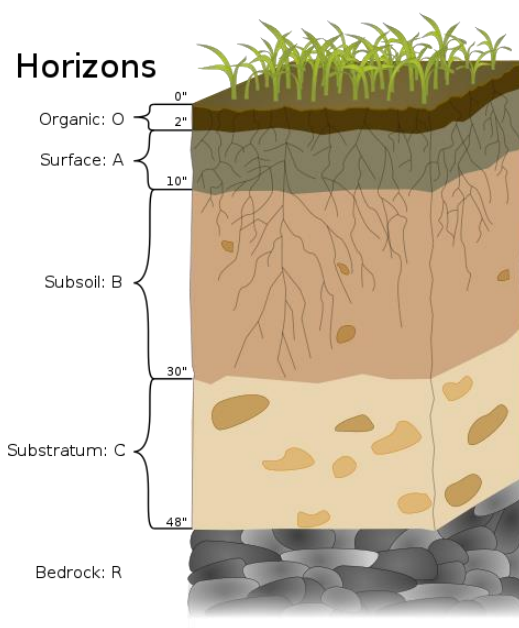
The soil surface is glistening and thus the timer should be stopped at this point

Collect Intact Soil Cores

1. Brush away surface litter and debris from the 2 coring areas spaced 1 meter from each other.
2. Remove litter layer that is greater than the size of a quarter, i.e. pine needles, large leaves, etc.



Do not remove the organic O horizon or uproot plants.



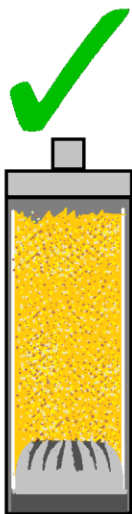
Example of organic soil horizon with a vegetation layer on top.

Image Credit: https://commons.wikimedia.org/wiki/File:Soil_Horizons.svg

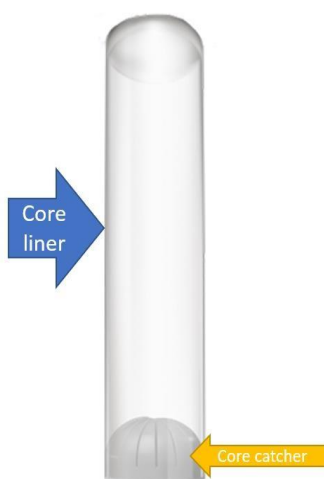
3. Locate a sample coring tube and remove the end caps.
4. Assemble the intact coring assembly by inserting the core catcher (white cup, if needed) into the base of the soil core liner (see attached photos).

When to use a core catcher

Sandy Soil



Clayey Soil



5. Place the core liner into the AMS intact coring barrel with the core catcher (if needed) at the bottom.



6. Screw the AMS core sampler cap onto the AMS intact coring barrel. Use the provided wrenches (strap and crescent).



7. Screw the corer onto the slide hammer using the two crescent wrenches provided. If more height is needed, you can add the 60 cm and 90 cm extension.



To prevent damage to the equipment, ensure that all connections are secure and do not force them together this can result in cross threading and damage to the equipment.

8. Hammer the soil corer into the ground by sliding the hammer up and striking down.



9. Hammer the soil corer until no further progress is made or until the top of the coring barrel is even with the surface soil.
10. Remove the core from the ground by reversing the slide hammer motion (sliding the hammer down and striking up).
- a. Alternatively, remove the slide hammer and attach the handle to turn the corer in the opposite direction and pull the core up.



- a. If you are still having trouble removing the core, firmly rock the core left and right to loosen the core from the surrounding soil and remove the core vertically by pulling upward.
- b. If you feel the soil will not stay in the core liner during removal, dig adjacent to the core, and insert a trowel horizontally at the bottom of the core prior to removal from the ground.



11. Remove the cap of the corer using the strap wrench around the coring barrel and the crescent wrench at the top.



12. Using a gloved hand, push from the base of the soil core upwards to push the liner up out of the top of the coring device. If the core is stuck, use the provided screwdriver to gently push it out.

13. If the core is a full 30 cm, pull the core liner out maintaining the vertical orientation and cap both the top and bottom of the core liner with the caps provided and seal with black/orange electrical tape.



If core doesn't fill 30 cm, trim the pool noodle using the provided utility knife to fill the remaining space in the coring tube.



14. Repeat steps 4-10 with the second tube. The first core sample should be your “Core A” (abiotic), and the second is “Core B” (biotic core). The core B should be taken approximately 1 meter from where the core A was sampled. Please label with provided stickers.
- a. Abiotic cores will be used to measure soil pore structures and hydraulic properties, it is imperative that the soil core remain as intact as possible.
15. **Label cores to indicate “top” and “bottom” and ensure that they are labelled with your NEXUS project ID and sampling event.**

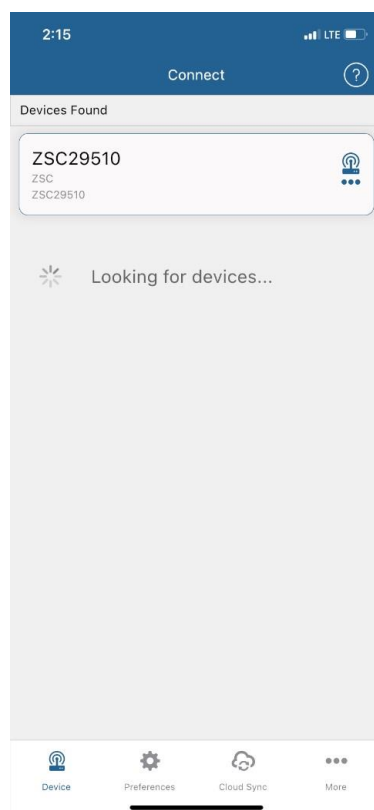
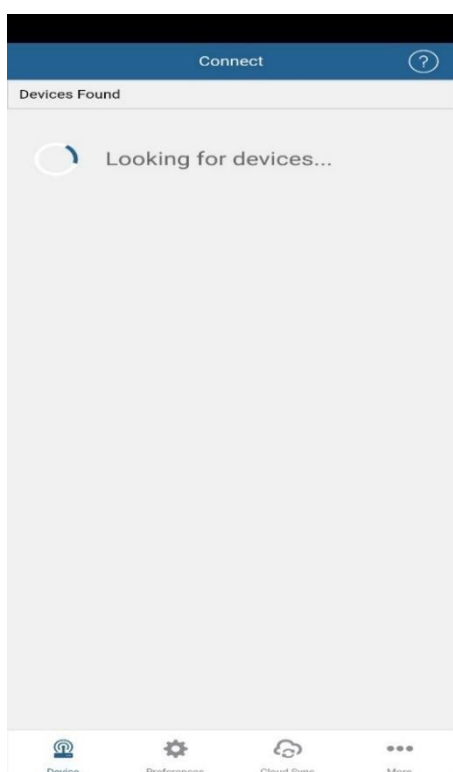


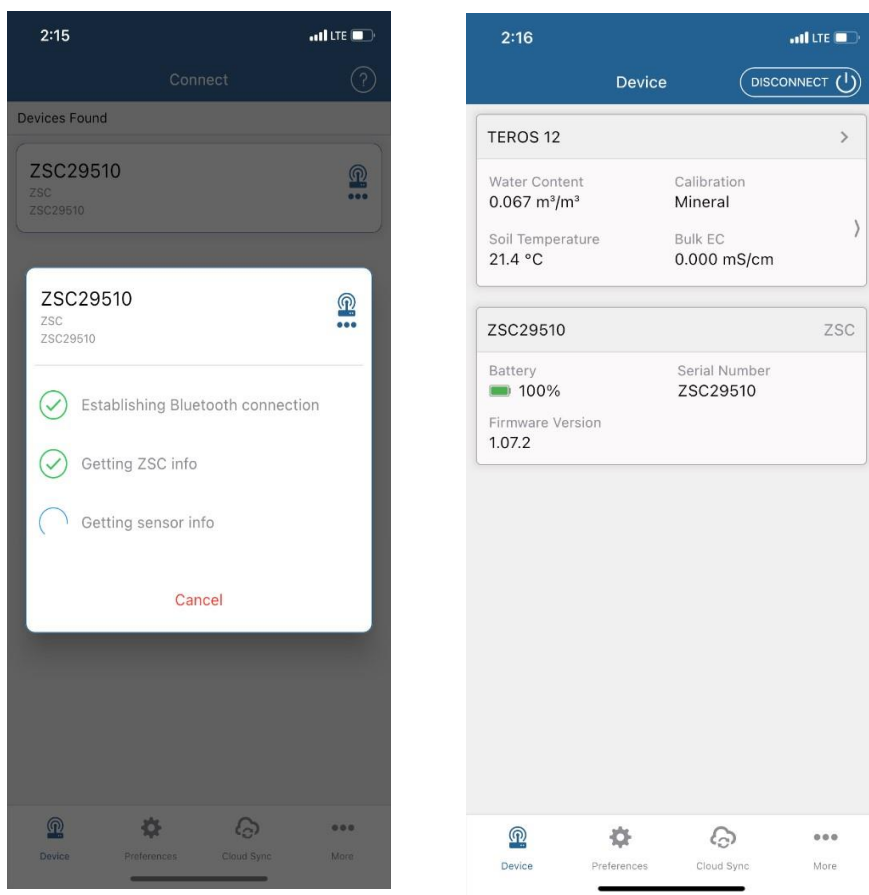
Collect Field Measurements

1. At the center point between the two soil cores, collect measurements for temperature, moisture, and electrical conductivity.
2. Insert the Teros 12 serial cable into the top port on the Bluetooth sensor.



3. Turn on the ZSC Bluetooth sensor by pressing down the button atop. Blue light will start blinking.
4. On your mobile device, enable Bluetooth and open the Zentra Utility app. App will automatically start scanning for **Meter Group ZSC Bluetooth sensors nearby**.
5. In the screen “Devices Found” connect the ZSC Bluetooth device by tapping the device that appears.

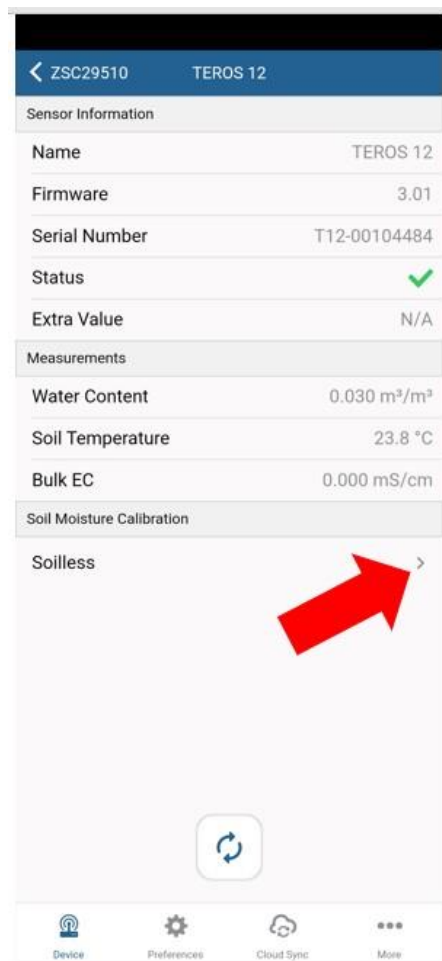
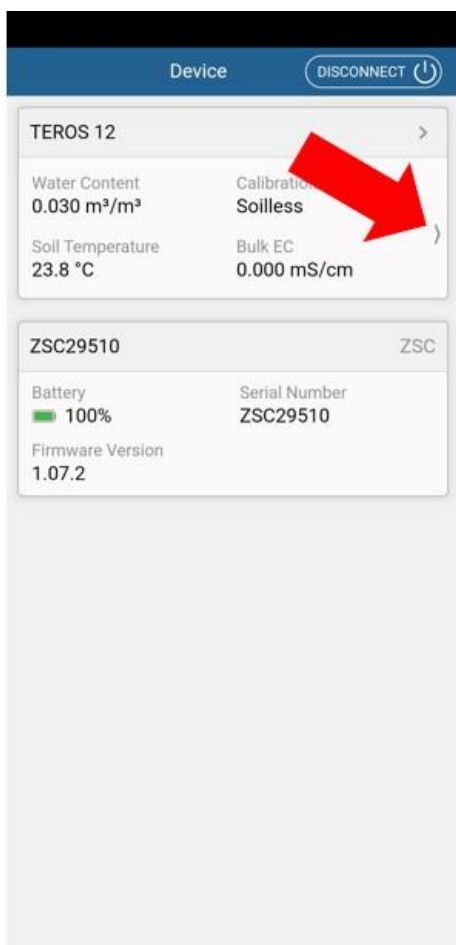




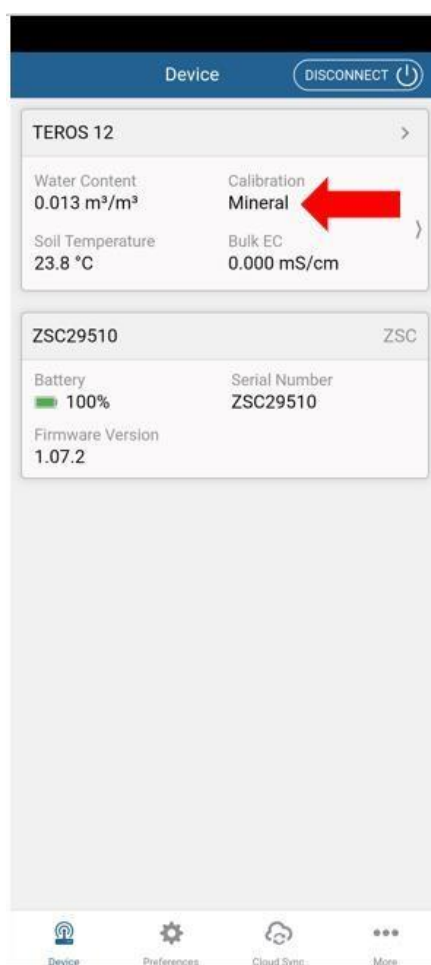
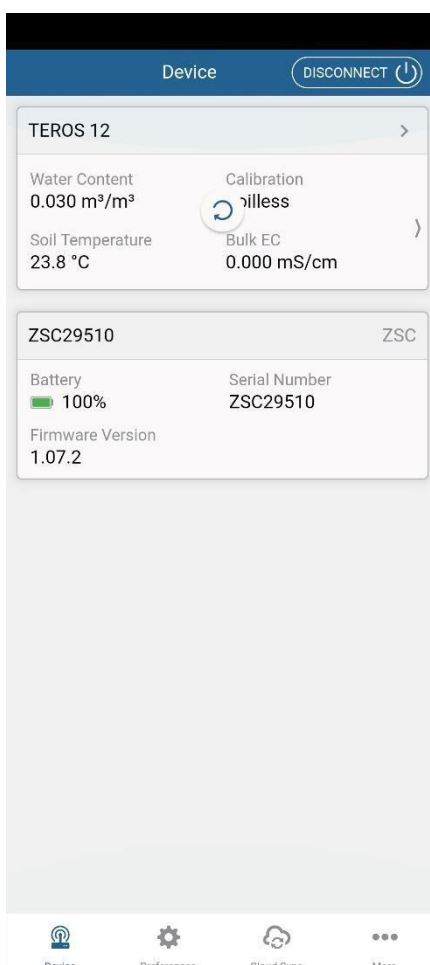
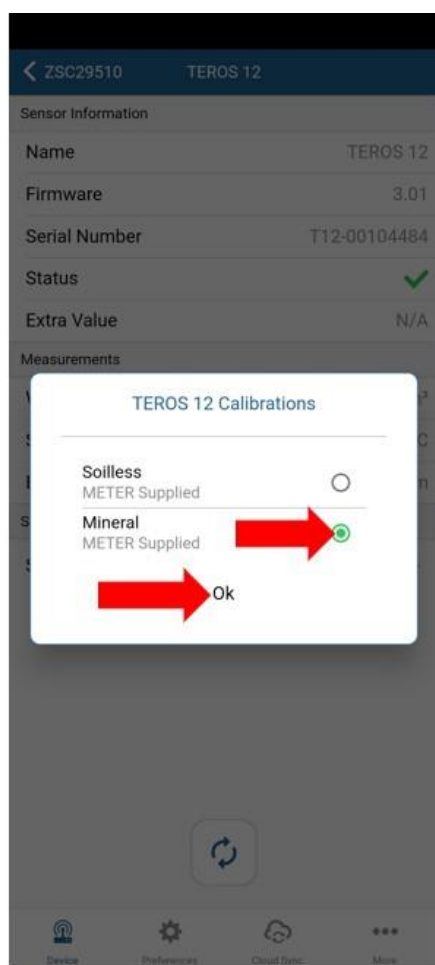
6. Under calibration On the Teros 12 section of the screen, ensure that the instrument calibration is set to Mineral.



7. If Teros 12 calibration is set up to Soilless, then follow steps bellow:



4 On TEROS 12 section, tap on arrow to the right. 5 Under **Soil Moisture Calibration**, tap on right arrow.



6 Select Mineral, then press OK. Go back to main screen and refresh app. Confirm Calibration is set up to Mineral.

8. Remove the foam cover over the 3 metal probes.



USE CAUTION PROBES ARE SHARP.



9. Gently press the probes into the ground. If there is any resistance, move the probe over a few centimeters and try again.



10. Once the probe is in the ground, allow the readings to stabilize for 3 minutes.



11. Record measurements on the metadata sheet.

Collect Surface Cores

1. To collect surface cores using the smaller 5 cm diameter cores, remove the litter layer from a spot within 1 m of Core A and Core B.
2. Place the coring liner into the soil and the black paddle across the top and use the mallet to hammer in the coring liners.
3. Use the trowel to clear ~1-2 cm deep section around the buried core to facilitate extraction.
4. Cap the top of the core liner (this forms a vacuum and reduces the amount of material lost) and rock the core back and forth to release it and pull the core liner and core up.
 - a. If the core cannot be easily removed use the shovel to dig the core out.



5. Cap the mini cores and wrap electrical tape around where the cap meets the core liner to seal the core.

6. Label caps to indicate “top” and “bottom” and ensure cores are labelled with your provided stickers.



7. Place the closed mini cores in the provided bag.



Alternative Core Collection Methods:



If an alternative method is required, XCT data will not be available as core structure is disturbed.

Collecting Auger Cores

1. If the above coring methods do not work for your soils, you can collect cores using the soil recovery auger (SRA).



Soil recovery auger



2. Place a soil core liner with the soil catcher in the bottom into the SRA and recap the auger.
3. Attach the SRA to the appropriate extender length and handle.
4. Using the provided mallet, hammer the corer into the soil about 2 cm to start the corer.
5. Once started turn the auger handle clockwise to collect a core. Stop once the auger can no longer core or the auger has reached the full depth.
6. Turn the auger the opposite direction and pull the core up.

Depth Stratified Method

1. Clear vegetation.
2. Use a trowel to discard the first couple scoops of soil.
3. Begin filling the core liner with soil. Be sure to capture the full 30 cm depth range.
4. Please indicate on the metadata sheet that intact soil cores failed, and in the notes section provide details on the coring alternative. I
5. If possible, maintain the approximate vertical structure of the core (e.g., mineral vs. organic layers). Note whether approximate spatial structuring was maintained on the metadata sheet.



Fill backfill holes to the best of your ability with surrounding surface soil to avoid leaving tripping hazards.

Checklist

Transport and Store Samples After Collection

1. After cores are collected, transport them on blue ice to a refrigerator (4 °C) and store vertically when possible. **Do not freeze cores.**

Upload Metadata and Photos

1. Upload site images to your project's folder on MONet Google drive.
2. Scan field metadata pages in order to send paper copies back with the soil cores.
3. Submit sample metadata on Science Central LIMS portal. Contact monet@pnnl.gov if you have issues submitting.

Shipping

1. Store cores vertically in the refrigerator until they are shipped between Monday and Wednesday. **It is critical that overnight shipments be made no later than Wednesday.**
2. Pack the cooler close to the time FedEx will ship the package. This maximizes the time cores stay cold.
3. When you're ready to ship, place each sample location's cores in the provided large zip top bags and place vertically in the cooler.
4. Place the metadata sheet into a plastic bag and seal it to prevent water damage. Place the frozen "blue ice" and other unused consumables back into the cooler and seal with tape.
5. Pack all remaining materials and the soil probes as received in the pelican case.
6. Please do not return used gloves or other waste generated. Dispose of the waste appropriately.
7. Drop the package off at FedEx, or have it picked up by FedEx (must ensure same day pick up).
8. Provide Fedex with the PNNL FedEx account number and shipping address provided by the project manager. Ask for "First Overnight" shipping.
9. **On the same day you ship the package, email the FedEx tracking number to monet@pnnl.gov.**

FIELD MEASUREMENTS						
Sample Set #	Sample Name	Infiltration Time 1 (mm:ss)	Infiltration Time 2 (mm:ss)	Bulk EC (mS/cm)	Soil Temp C	Vol. Water Content (m3/m3)
EXAMPLE	60001_1	03:12 mm:ss	7:48 mm:ss	0.015 mS/cm	14.5 C	0.294 m3/m3
1						
2						
3						
4						
5						
6						
7						
8						
9						

Sample Set #	Sample Name	Collection Time (HH:MM)	Latitude, Longitude (4 decimal places)	Depth (cm)	Device (corer, auger, hand)	Notes
EXAMPLE	60001_1_A	03:12 mm:ss	46.3486, 119.2777	0-30 cm	Corer	Very compacted
	60001_1_B			0-29 cm	Auger	
	60001_1_C1-4			0-10 cm	Hand	
1						
2						
3						

Sample Set #	Sample Name	Collection Time (HH:MM)	Latitude, Longitude (4 decimal places)	Depth (cm)	Device (corer, auger, hand)	Notes
4						
5						
6						
7						

Sample Set #	Sample Name	Collection Time (HH:MM)	Latitude, Longitude (4 decimal places)	Depth (cm)	Device (corer, auger, hand)	Notes
8						
9						